

Non-Functional Requirements (NFR)

E-commerce & Campaign Profitability Analyzer

Non-functional requirements define the quality, reliability, integrity, and maintainability expectations for the analytical system.

Requirements

NFR-2.1 - Financial Precision

Requirement	Monetary metrics must be stored using appropriate numeric precision.
Specification	Financial values use NUMERIC(10,2) or another appropriate precision.
Reason	Financial calculations require consistent decimal precision and should avoid inappropriate rounding.

NFR-2.2 - Performance

Requirement	Analytical SQL queries should have a defined execution-time target on a representative dataset.
Specification	Testing documents: dataset size, environment, query tested, execution time, test conditions.
Benchmark	Not yet recorded - to be measured against the project's ~1,445-order dataset in MySQL and logged here before final submission. The project does not claim a performance benchmark unless it has actually been tested.

NFR-2.3 - Null & Zero Handling

Requirement	The system must handle NULL and zero values appropriately.
Specification	For ROAS: if Ad Spend = 0, ROAS = NULL. Prevents division-by-zero errors and avoids an artificial profitability value.

NFR-2.4 - Data Integrity

Requirement	The database must protect relational consistency.
Specification	Primary keys, foreign keys, NOT NULL constraints, valid status checks, appropriate data types, quantity validation (e.g. Order Item Quantity > 0).

NFR-2.5 - Auditability & Traceability

Requirement	Business metrics must be traceable from the original business requirement through to the final analytical output.
Specification	Chain: Requirement → Business Rule → SQL Logic → Test Case → Reporting Output. A stakeholder can trace why a metric exists, which rule defines it, how it's calculated, how it was tested, and where it appears in reporting.

NFR-2.6 - Data Quality

Requirement	The system should support checks for data consistency issues.
Specification	Duplicate records, orphan records, invalid statuses, missing required fields, payment reconciliation differences, refund inconsistencies, referential integrity issues.

NFR-2.7 - Maintainability

Requirement	SQL and documentation should be structured so business rules and calculations can be understood and maintained by another analyst.
Specification	Clear naming conventions, modular SQL scripts, CTEs for multi-step calculations, separate validation queries, documented business rules, consistent metric definitions.

NFR Summary

ID	Requirement	Key Expectation
NFR-2.1	Financial Precision	Appropriate numeric precision
NFR-2.2	Performance	Actual benchmark must be documented
NFR-2.3	Null Handling	Zero-spend ROAS returns NULL
NFR-2.4	Data Integrity	Keys, constraints and valid statuses
NFR-2.5	Auditability	Requirement-to-output traceability

ID	Requirement	Key Expectation
NFR-2.6	Data Quality	Validation of data consistency
NFR-2.7	Maintainability	Clear, modular and documented implementation

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